

Course Outlines

- Introduction to Project Management
- Technology to Project Management
- Project Scheduling
- Risk Management
- Change Management
- Leadership and Ethics in Project
- Project Implementation
- Closure and Evaluation

Topic One:

WHAT IS PROJECT?

A project is “a temporary endeavor undertaken to create a unique product, service, or result.” Project in general refers to a new endeavor with specific objective and varies so widely that it is very difficult to precisely define it. Project is a unique process, consist of a set of coordinated and controlled activities with start and finish dates, undertaken to achieve an objective conforming to specific requirements, including the constraints of time cost and resource.

A project is an activity that:

- is temporary having a start and end date
- is unique
- brings about change
- has unknown elements, which therefore create risk
- Generally projects are formed to solve a problem or take advantage of an opportunity.
- Business as usual activities can often be mistaken for projects.
- Generally it is the uniqueness of the activity that is the deciding factor – do we do this every year? If so, then it is not truly a project – although you can use project methods to get it done.

EXAMPLES OF IT PROJECTS

Projects can be large or small and involve one person or thousands of people. They can be done in one day or take years to complete. As described earlier, IT projects involve using hardware, software, and networks to create a product, service, or result. Examples of IT projects include the following:

A large network of healthcare providers updates its information systems and procedures to reduce hospital acquired diseases.

- A team of students creates a smartphone application and sells it online.
- A company develops a driverless car.
- A college upgrades its technology infrastructure to provide wireless Internet access across the whole campus as well as online access to all academic and student service information.
- A company implements a new system to increase sales force productivity and customer relationship management that will work on various laptops, smartphones, and tablets.
- A television network implements a system to allow viewers to vote for contestants and provide other feedback on programs via social media sites.
- A government group develops a system to track child immunizations.
- A large group of volunteers from organizations throughout the world develops standards for environmentally friendly or green IT.

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- A global bank acquires other financial institutions and needs to consolidate systems and procedures.
- Government regulations require monitoring of pollutants in air and water.
- A multinational firm decides to consolidate its information systems into an integrated enterprise resource management approach.

WHY DO PROJECT FAIL?

1. Poor project and program management discipline
2. Lack of executive-level support
3. Wrong team members
4. Poor communication
5. No measures for evaluating the success of the project
6. No risk management
7. Inability to manage change

PROJECT CHARACTERISTICS

Despite above diversities, projects share the following common characteristics.

- ✓ Unique in nature.
- ✓ Have definite objectives (goals) to achieve.
- ✓ Requires set of resources.
- ✓ Have a specific time frame for completion with a definite start and finish.
- ✓ Involves risk and uncertainty.
- ✓ Requires cross-functional teams and interdisciplinary approach
- ✓ A project is temporary
- ✓ A project drives change and enables value creation
- ✓ A project is developed using progressive elaboration.
- ✓ A project requires resources, often from various areas
- ✓ A project should have a primary customer or sponsor.

PROJECT SUCCESS/BENEFITS

1. The project met scope, time, and cost goals
2. The project satisfied the customer/sponsor
3. The results of the project met its main objective, such as making or saving a certain amount of money, providing a good return on investment, or simply making the sponsors happy

PROJECT CONSTRAINTS

1. **Scope:** What work will be done as part of the project? What unique product, service, or result does the customer or sponsor expect from the project? How will the scope be verified?
 2. **Time:** How long should it take to complete the project? What is the project's schedule? How will the team track actual schedule performance? Who can approve changes to the schedule?
 3. **Cost:** What should it cost to complete the project? What is the project's budget? How will costs be tracked? Who can authorize changes to the budget?
- Project scope – Have all the project requirements (i.e., deliverables) been completed?
 - Project cost – Is the cost of the project close to the amount the customer has agreed to pay?
 - Schedule – Was the project completed on time?
 - Customer satisfaction – Is the customer happy with the quality of the project

Successful project management means meeting all three goals (scope, time, and cost)—and satisfying the project's sponsor

WHO IS PROJECT MANAGER?

Project Managers are essentially jugglers. They must make sure that everything keeps to task, that potential issues are quickly eliminated and the project is delivered on time, all the while making sure everyone knows what is happening and the project quality and budget are acceptable. Specifically they:

- ✓ direct all activities required to successfully meet the project objectives
- ✓ manage risk – scanning ahead for potential issues and resolving them before they become a problem
- ✓ solve problems - recommending alternative approaches to problems that arise and providing guidance to the Project Sponsor
- ✓ track and report project progress
- ✓ communicate to all stakeholders in the project

A project manager must:

A person with a diverse set of skills – management, leadership, technical, conflict management, and customer relationship –who is responsible for:

- initiating,
- planning,
- executing,
- controlling,
- monitoring,
- and closing down a project.

A program manager:

- Provides leadership and direction for the project managers heading the projects within a program.
- It also coordinate the efforts of project teams, functional groups, suppliers, and operations staff supporting the projects to ensure that products and processes are implemented to maximize benefits.
- They are responsible for more than the delivery of project results; they are change agents responsible for the success of products and processes developed by those projects.
- They often have review meetings with all their project managers to share important information and coordinate important aspects of each project. Many program
- They managers worked as project managers earlier in their careers, and they enjoy sharing their wisdom and expertise with their project managers. Effective program managers
- They recognize that managing a program is much more complex than managing a single project.
- They recognize that technical and project management skills are not enough—program managers must also possess strong business knowledge, leadership capabilities, and communication skills.

SIX TRAITS OF HIGHLY EFFECTIVE PROJECT MANAGERS AS FOLLOWS:

1. Be a strategic business partner.
2. Encourage and recognize valuable contributions.
3. Respect and motivate stakeholders.
4. Be fully vested in success.
5. Stress integrity and accountability
6. Work in the gray/Be able to deal with ambiguity.

ROLES AND RESPONSIBILITIES OF PROJECT MANAGER

The basic role & Responsibility of a Project manager are as under:

1. Organize and implement diverse and scattered activities to achieve time, cost and performance goals.

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2. Develop project execution plan.
3. He should inculcate enthusiasm, team spirit, confidence and reputations for excellence.
4. Develop systems and procedure for accomplishment of project objectives.
5. Direct, coordinate and control the project activities.
6. Guiding and Problem solving.
7. Satisfaction of customers, government and public.
8. He should be an entrepreneur.

WHAT IS PROJECT MANAGEMENT?

Project management is “the application of knowledge, skills, tools, and techniques to project activities to meet project requirements. Project management is the discipline of organizing and managing resources in such a way that these resources deliver all the work required to complete a project within defined scope, time, and cost constraints

Project management is a tool of management for planning and control. The objective of which has been defined as an attempt to make the most efficient and effective use of resources, viz.: manpower, equipment, facilities, materials, money, technology and information, so that the organization’s objectives and goals can be achieved within budget, on schedule and at the desired performance/technology level, while adhering to the ever changing environmental input factors, like legal, social, political, economical and technological.

Project management can be defined as the planning, organizing, directing and controlling of company resources for a relatively short-term objective that has been established to complete specific goals and objectives.

PROJECT STAKEHOLDERS

Stakeholders are the people involved in or affected by project activities, and include the project sponsor, project team, support staff, customers, users, suppliers, and even opponents of the project

KEY ELEMENTS OF PROJECT MANAGEMENT

- Scope
- Time
- Cost
- Resource

(1) Scope

This is your project’s DNA – a crystal-clear understanding of what’s included (and excluded). It’s more like the boundary of the project. It outlines what will be delivered, detailing the tasks, deliverables, and boundaries. Defining the scope involves understanding the project’s objectives and breaking them down into manageable tasks. A well-managed scope is crucial to avoid scope creep that can inflate costs and deadlines.

(2) Time

A project timeline is a schedule of activities, milestones, and deliverables. It’s a visual representation, often in the form of a Gantt chart that helps everyone understand the sequence of tasks and their dependencies. Developing a timeline requires estimating the duration of tasks, sequencing them, and allocating resources. It includes several steps, including – Identifying tasks & dependencies: List all the tasks involved in the project and identify any dependencies between them. Estimate task durations: For each

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task, estimate how long it will take to complete, considering its complexity, the skills of the assigned resource, and potential risks.

Sequencing & scheduling: Use your task list and dependencies to determine the order in which tasks need to be completed. There might be some flexibility, so consider factors like resource availability and task criticality when creating your schedule.

Read More: [How to Build a Project Timeline in 6 Simple Steps](#)

(3) Budget

This involves predicting the costs associated with the project. It's not just about the initial estimates but about managing costs throughout the project lifecycle. Estimate project costs meticulously, including labor, materials, and potential contingencies. Track expenses throughout the project to identify deviations and ensure you stay within budget. One of the most common ways to manage budgets is to maintain a spreadsheet. But, in my experience, that often becomes a tedious task for complex projects.

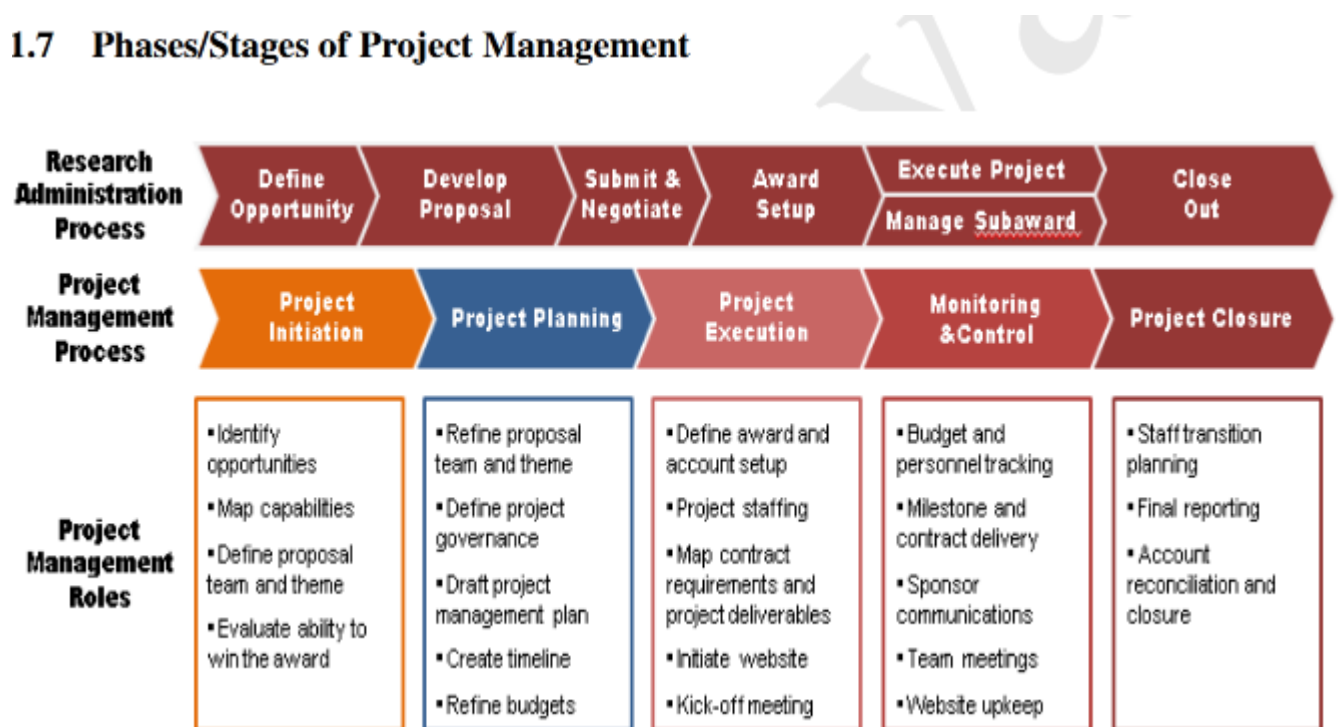
(4) Resource

Your project is only as strong as its team. And the team is only as strong as the available resources are. Resource allocation involves assigning the most suitable resources to the right tasks at the right time. It involves balancing the availability of team members, equipment, and materials with the project's needs, ensuring that resources are used efficiently. Proper resource management also ensures everyone's workload is balanced.

(5) Procurement Management

As the name suggests, procurement management is the process of acquiring goods and services from external sources. For a manufacturing project, procurement management would include negotiating with suppliers for raw materials, ensuring timely delivery, and managing contracts to ensure compliance with project timelines and budgets.

1.7 Phases/Stages of Project Management



PROJECT MANAGEMENT TOOLS AND TECHNIQUES

Some of the tools and techniques used in project are:

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- Data gathering: benchmarking, brainstorming, check sheets, checklists, focus groups, interviews, market research, questionnaires and surveys, and statistical sampling
- Data analysis: alternatives analysis, assessment of other risk parameters, assumption and constraint analysis, cost of quality, cost-benefit analysis, decision tree analysis, document analysis, earned value analysis, and several other tools fit in this category

TOOLS & TECHNIQUES FOR PROJECT MANAGEMENT

Following are the important tools and techniques for effective project management.

1. Project selection techniques.
 - (a) Cost-benefit analysis.
 - (b) Risk and sensitivity analysis.
2. Project implementation (execution) planning techniques.
 - (a) Work break down structure (WBS)
 - (b) Project implementation plan.
 - (c) Project responsibility matrix.
 - (d) Project management manuals.
3. Project scheduling techniques.
 - (a) Line of Balance (LOB)
 - (b) Bar Charts
 - (c) Network Techniques (CPM/PERT)
4. Project monitoring techniques.
 - (a) Progress measurement technique.
 - (b) Performance monitoring technique.
 - (c) Updating, reviewing and reporting technique.
5. Project cost and productivity techniques
 - (a) Value engineering.
 - (b) Budgetary control.
 - (c) Cost/WBS.
6. Project communication techniques.
 - (a) Computerized information system.
 - (b) Control room.
 - (c) MIS

COMMON PROJECT MANAGEMENT TOOLS AND TECHNIQUES BY KNOWLEDGE AREA

Knowledge Area/Category	Tools and Techniques	Super Tools
Integration management	Project selection methods Project management methodologies Stakeholder analyses Work requests Project charters Project management plans Change control boards Project review meetings	Project management software Change requests Lessons-learned reports
Scope management	Statements of work Scope management plans Scope verification techniques Scope change controls	Scope statements Work breakdown structures Requirements
	Scope verification techniques Scope change controls	structures Requirements analyses
Schedule management	Project network diagrams Critical path analysis Crashing Fast tracking Schedule performance measurements	Gantt charts
Cost management	Project budgets Net present value Return on investment Payback analysis Earned value management Project portfolio management Cost estimates	
Quality management	Quality metrics Checklists Quality control charts Pareto diagrams Fishbone diagrams Maturity models Statistical methods Test plans	

TABLE 1-1 Common project management tools and techniques by knowledge area (*continued*)

Knowledge Area/Category	Tools and Techniques	Super Tools
<i>Resource management</i>	Motivation techniques Empathic listening Responsibility assignment matrices Project organizational charts Resource histograms Team building exercises	
<i>Communications management</i>	Communications management plans Conflict management Communications media selection Status reports Virtual communications Team building exercises	Kick-off meetings Progress reports
<i>Risk management</i>	Risk management plans Risk registers Probability/impact matrices Risk rankings	
<i>Procurement management</i>	Make-or-buy analyses Contracts Requests for proposals or quotes Source selections Supplier evaluation matrices	

- Data representation: affinity diagrams, cause-and-effect diagrams, control charts, flow charts, hierarchical charts, histograms, logical data models, matrix diagrams, matrix-based charts, mind mapping, probability and impact matrix, scatter diagrams, stakeholder engagement assessment matrix, stakeholder mapping/representation, and text-oriented formats
- Decision making: multi-criteria decision analysis and voting
- Communication: feedback and presentations
- Interpersonal and team skills: active listening, communication styles assessment, conflict management, cultural awareness, decision making, emotional intelligence, facilitation, influencing, leadership, meeting management, motivation, negotiation, networking, nominal group, observation/conversation, political awareness, team building
- Ungrouped: several other tools fit in this category

PROGRAM AND PROJECT PORTFOLIO MANAGEMENT

WHAT IS A PROGRAM?

A program is “a group of related projects, subsidiary programs, and program activities managed in a coordinated manner to obtain benefits and control not available from managing them individually

The following are examples of common programs in the IT field.

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- **Infrastructure:** An IT department often has a program for IT infrastructure projects. This program could encompass several projects, such as providing more wireless Internet access, upgrading hardware and software, enhancing computer security, and developing and maintaining corporate standards for IT.
- **Applications development:** This program could include several projects, such as updating an enterprise resource planning (ERP) system, purchasing a new off-the-shelf billing system, or developing a new capability for a customer relationship management system.
- **User support:** In addition to the many operational tasks related to user support, many IT departments have several projects to support users. For example, a project might provide a better e-mail system or develop technical training for users.

Topic Two:

IMPORTANCE OF LEADERSHIP SKILLS

1. **Technical project management skills:** Understanding the knowledge areas, process groups, and project management tools and techniques fall into this category.
2. **Strategic and business management skills:** Topics include strategic planning and financial management (described in more detail in Chapter 4), accounting, marketing, and other topics listed in the previous section.
3. **Leadership skills:** Leadership and management are terms often used interchangeably, although there are differences. Generally, a leader focuses on long-term goals and big-picture objectives, while inspiring people to reach those goals. A manager often deals with the day-to-day details of meeting specific goals. Some people say that, “Managers do things right, and leaders do the right things.” “Leaders determine the vision, and managers achieve the vision.” “You lead people and manage things.”

PMBOK® Guide – Sixth Edition briefly describes the following Leadership styles:

1. **Laissez-faire:** Meaning “let go,” this hands-off approach lets teams determine their own goals and how to achieve them.
2. **Transactional:** This management by exception approach focuses on achieving goals or compliance by offering team members appropriate rewards and punishments.
3. **Servant leader:** People using this approach focus on relationships and community first and leadership is secondary.
4. **Transformational:** By working with others to identify needed changes, these leaders empower others and guide changes through inspiration.
5. **Charismatic:** These people can inspire others based on their enthusiasm and confidence.
6. **Interactional:** This leadership style is a combination of transactional, transformational, and charismatic.

Daniel Goleman, author of Emotional Intelligence, also wrote a book called Primal Leadership, which describes six different styles of leadership and situations where they are most appropriate:

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1. **Visionary:** Needed when an organization needs a new direction, and the goal is to move people toward a new set of shared dreams. The leader articulates where the group is going, but lets them decide how to get there by being free to innovate, experiment, and take calculated risks.
2. **Coaching:** One-on-one style that focuses on developing individuals, showing them how to improve their performance. This approach works best with workers who show initiative and request assistance.
3. **Affiliative:** Emphasizes the importance of teamwork and creating harmony by connecting people to each other. This approach is effective when trying to increase morale, improve communication, or repair broken trust.
4. **Democratic:** Focuses on people's knowledge and skills and creates a commitment to reaching shared goals. This leadership style works best when the leader needs the collective wisdom of the group to decide on the best direction to take for the organization.
5. **Pacesetting:** Used to set high standards for performance. The leader wants work to be done better and faster and expects everyone to put forth their best effort.
6. **Commanding:** Most often used, also called autocratic or military style leadership. This style is most effective in a crisis or when a turnaround is needed.

ETHICS IN PROJECT MANAGEMENT

Ethics, loosely defined, is a set of principles that guides decision making based on personal values of what is considered right and wrong. Ethics guide us in making these types of decisions

2.2.1 We make decisions and take actions based on the best interests of society, public safety, and the environment.

2.2.2 We accept only those assignments that are consistent with our background experience, skills, and qualifications.

2.2.3 We fulfill the commitments that we undertake—we do what we say we will do.

3.2.1 We inform ourselves about the norms and customs of others and avoid engaging in behaviors they might consider disrespectful.

3.2.2 We listen to others' points of view, seeking to understand them.

3.2.3 We approach directly those persons with whom we have a conflict or disagreement.

4.2.1 We demonstrate transparency in our decision-making process.

4.2.2 We constantly reexamine our impartiality and objectivity, taking corrective action as appropriate.

4.3.1 We proactively and fully disclose any real or potential conflicts of interest to appropriate stakeholders.

5.2.1 We earnestly seek to understand the truth.

Project management software tools can be divided into three general categories based on functionality and price:

1. Low-end tools
2. Midrange tools
3. High-end tools

Topic Three:

INTRODUCTION TO CPM & PERT TECHNIQUE

Program Evaluation and Review Technique (PERT) and Critical Path Method (CPM) are two most widely used techniques in project management. The objectives of project management can be described in terms of a successful completion of project on time within the budgeted cost and adhering to technical specifications which satisfy the end users. A project is any human undertaking with a clear beginning and a clear ending.

INTRODUCTION TO CRITICAL PATH METHOD (CPM)

CPM is applicable to both large and small projects, taking from space programmes to wedding or horse shows. It is widely recognized and is the most versatile and potent management planning technique.

CPM is a technique, used for planning and controlling the most logical and economic sequence of operations for accomplishing a project. The project is analyzed into different activities whose relationship, as in PERT, are shown on the network diagram. The network is then utilized for optimising the use of resources, progress and control.

Introduction to Programme Evaluation Review Technique (PERT)

PERT was first used in 1957 as a method of planning and controlling the Polaris Missile Programme by Booz, Allen and Hamilton together with U.S. Naval department. The aim was to finish the project two years in advance. PERT describes basic network technique which includes planning, monitoring and control of projects.

PROGRAMME EVALUATION AND REVIEW TECHNIQUE (PERT)

The PERT technique is a method of minimizing trouble spots, programme bottlenecks, delays and interruptions by determining critical activities before they occur so that various activities in the project can be coordinated

- PERT is a very important technique in the field of project management. It is commonly employed for conducting the initial review of a project.
- PERT, actually, developed as a research and development planning tool where activity timings could not be estimated with enough certainty.
- PERT can be employed at those places where a project can not be easily defined in terms of time or resources required.
- PERT is mainly concerned with event and thus it is an event-oriented system.
- The basic tools used in PERT technique is the network or flow plan. Network consists of series of related events and activities

DIFFERENCE BETWEEN PERT & CPM

There is not much difference between PERT & CPM. In fact both were developed simultaneously and independently for the similar requirements. The only difference lies in the 'event' or 'activity' on which

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network is constructed. The 'activity' stands for the time-consuming parts of the project. It represents a job; on the other hand, an 'event', also called 'node' either beginning or an end of a job. The CPM analysis is activity-oriented and PERT is event-oriented.

1. The CPM is a kind of technique commonly known for its use in constructions project; whereas PERT mostly used in R & D projects like the projects considered being a kind of non-repetitive nature.
2. Both techniques have different concept of working like CPM uses deterministic concept for its work whereas probabilistic concept is used by PERT.
3. Networking approach also varies in both techniques like CPM uses networking that based upon activity oriented; on contrary PERT uses networking that based on event.
4. In PERT technique, an estimation of time for different activities is not perfect and accurate whereas in CPM activities duration are estimated with the quality of accuracy.
5. Both of the techniques used for same purpose but the nature of their working is different and some can do their work efficiently with CPM and some can do same with PERT.
6. The most important thing to solute is to minimize time so that they can get good result for their cost factor. In PERT, time is said to be a controlling factor.
7. CPM has one-time estimate while PERT has three time estimates.
8. Unpredicted activities are managed with the help of Program Evaluation and Review Technique (PERT), whereas CPM is used for dealing those projects that have some alarmed activities. These two techniques said to be a key element for the management of any project.

ADVANTAGES OF PERT

1. It determines the expected duration of activities and consequently of the project duration.
2. It helps the management in handling the uncertainties involved in the project and thus reduces the risk element in the object.
3. In enables the management to make optimum allocation of limited resources.
4. It process for the right action, at the right point and the right times in the organization.
5. It determines the most economical schedule for fixed project duration.

DISADVANTAGES OF PERT

1. The time estimates to perform activities constitutes a major limitation of this technique. If the estimates are not satisfactory then the network will be highly unrealistic.
2. The probability distribution of total time is assured to be normal, which in real life situation may not be true.

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3. The simple PERT technique does not consider the resources required at various stages of the project. If a certain resource may be used to perform more than one activity and at the same time, it can be used for only one activity at a time, then the network diagram will become infeasible.

4. Use of this technique for active control of a project requires frequent updating and revision of PERT calculations and this proves quite a costly affair

ADVANTAGES OF CPM

1. It helps in ascertaining the time schedule of activities having sequential relationships.
2. It makes the control function of the management.
3. It makes better and detail planning possible.
4. It identifies the most critical elements in the project. Thus, the management is kept alert and prepared to pay due attention to the critical activities of the project.

DISADVANTAGES OF CPM

1. CPM fails to incorporate statistical analysis in determining the time estimates.
2. It operates on the assumption that there is a precise known time that each activity in the project will take. It may not be true in reality.
3. It cannot be used as a controlling device since any change introduced will change the entire structure of the network. Thus, CPM cannot be used as a dynamic controlling device.

PROJECT FORMULATION

Project Formulation is the processes of presenting a project idea in a form in which it can be subjected to comparative appraisals for the purpose of determining in definitive terms the priority that should be attached to a project under sever resource constraints. Project Formulation involves the following steps (Figure 1).

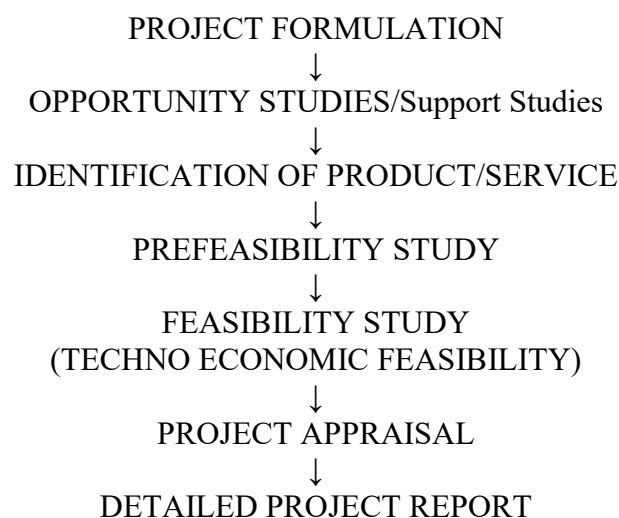


Figure 1.: Project Formulation –Schematic view

OPPORTUNITY STUDIES

An opportunity study identifies investment opportunities and is normally undertaken at macro level by agencies involved in economic planning and development. In general opportunity studies there are three types of study – Area Study, sectoral and Sub-sectoral Studies and Resource Based Studies. Opportunity Studies and Support studies provide sound basis for project identification.

PRE FEASIBILITY STUDIES / OPPORTUNITY STUDIES

A pre-feasibility study should be viewed as an intermediate stage between a project opportunity study and a detailed feasibility study, the difference being primarily the extent of details of the information obtained. It is the process of gathering facts and opinions pertaining to the project. This information is then vetted for the purpose of tentatively determining whether the project idea is worth pursuing furthering. Pre feasibility study lays stress on assessing market potential, magnitude of investment, , technical feasibility, financial analysis, risk analysis etc. The breadth and depth of pre feasibility depend upon the time available and the confidence of the decision maker. Pre feasibility studies help in preparing a project profile for presentation to various stakeholders including funding agencies to solicit their support to the project. It also throws light on aspects of the project that are critical in nature and necessitate further investigation through functional support studies.

FEASIBILITY STUDY

Feasibility Study forms the backbone of Project Formulation and presents a balanced picture incorporating all aspects of possible concern. The study investigates practicalities, ways of achieving objectives, strategy options, methodology, and predict likely outcome, risk and the consequences of each course of action. It becomes the foundation on which project definition and rationale will be based so that the quality is reflected in subsequent project activity. A well conducted study provides a sound base for decisions, clarifications of objectives, logical planning, minimal risk, and a successful cost effective project. Assessing feasibility of a proposal requires understanding of the STEEP factors. These are as under Social, Technological, Ecological, Economic, and Political.

A feasibility study is not an end in itself but only a means to arrive at an investment decision. The preparation of a feasibility study report is often made difficult by the number of alternatives (regarding the choice of technology, plant capacity, location, financing etc.) and assumptions on which the decisions are made. The project feasibility studies focus on

- Economic and Market Analysis
- Technical Analysis
- Market Analysis
- Financial Analysis
- Economic Benefits
- Project Risk and Uncertainty
- Management Aspects

PROJECT APPRAISAL

The project appraisal is the process of critical examination and analysis of the proposal in totality. The appraisal goes beyond the analysis presented in the feasibility report. At this stage, if required compilation of additional information and further analysis of project dimensions are undertaken. At the end of the process an appraisal note is prepared for facilitating decision on the project implementation.

The appraisal process generally concentrates on the following aspects.

- Market Appraisal: Focusing on demand projections, adequacy of marketing infrastructure and competence of the key marketing personnel.
- Technical Appraisal: Covering product mix, Capacity, Process of manufacture engineering know-how and technical collaboration, Raw materials and consumables, Location and site, Building, Plant and equipments, Manpower requirements and Break-even point.
- Environmental Appraisal: Impact on land use and micro-environment, commitment of natural resources, and Government policy.

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- Financial Appraisal: Capital, rate of return, specifications, contingencies, cost projection, capacity utilization, and financing pattern.
- Economic Appraisal: Considered as a supportive appraisal it reviews economic rate of return, effective rate of protection and domestic resource cost.
- Managerial Appraisal: Focuses on promoters, organization structure, managerial personnel, and HR management.
- Social Cost Benefit Analysis (SCBA): Social Cost Benefit Analysis is a methodology for evaluating projects from the social point of view and focuses on social cost and benefits of a project. There often tend to differ from the costs incurred in monetary terms and benefits earned in monetary terms by the project SCBA may be based on UNIDO method or the Little-Mirriles (L-M) approach. Under UNIDO method the net benefits of the project are considered in terms of economic (efficiency) prices also referred to as shadow prices. As per the L-M approach the outputs and inputs of a project are classified into (1) traded goods and services (2) Non traded goods and services; and (3) Labor. All over the world including India currently the focus is on Economic Rate of Return (ERR) based on SCBA assume importance in project formulation and investment decisions

DETAILED PROJECT REPORT (DPR)

Once the projects are appraised and the investment decisions are made a Detailed Project Report (DPR) is prepared. It provides all the relevant details including design drawings, specifications, detailed cost estimates etc. and this would act as a blue print for project implementation.

PROJECT MANAGEMENT TECHNIQUES

These project management techniques can be classified under two broad categories i.e., Bar Charts and Networks.

Bar Charts

Bar charts are the pictorial representation of various tasks required to be performed for accomplishment of the project objectives. These charts have formed the basis of development of many other project management techniques.

Gantt Chart

Henry L Gantt (1861 – 1919) around 1917 developed a system of bar charts for scheduling and reporting progress of a project. These charts latter were known as Gantt Charts. It is a pictorial representation specifying the start and finish time for various tasks to be performed in a project on a horizontal time-scale

Milestone Chart

Milestone chart is an improvement over the bar chart (Gantt chart) by introducing the concept of milestone.

Networks

The network is a logical extension of Gantt's milestone chart incorporating the modifications so as to illustrate interrelationship between and among all the milestones in an entire project

PROJECT LIFE CYCLE

A project life cycle is a collection of project phases. Projects pass through these phases from their start to their completion. The PMBOK® Guide – Sixth Edition describes the generic life cycle to include the following four phases:

1. Starting the project
2. Organizing and preparing

3. Carrying out the work
4. Finishing the project

PRODUCT LIFE CYCLES

1. Predictive life cycle: The scope, schedule, and cost are determined early, and changes to scope are carefully managed. PMI also refers to predictive life cycles as waterfall.
2. Iterative life cycle: The scope is determined early, but time and cost estimates are modified as the understanding of the product increases. Iterations are used to develop the product through a series of repeated cycles to add to the functionality of the product. This approach works best when there is a high degree of change and a low frequency of delivery.
3. Incremental life cycle: Deliverables are produced through a series of iterations that add functionality within a set time frame. The deliverable is not complete until after the final iteration. This approach works best when there is a low degree of change and a high frequency of delivery.
4. Adaptive life cycle: Stakeholders define and approve the detailed scope before the start of an iteration, producing a useable product at the end of each iteration. PMI also refers to adaptive life cycles as agile or change- driven. This approach works best when there is a high degree of change and a high frequency of delivery.
5. Hybrid life cycle: A combination of approaches is used based on the nature of the work. For example, some deliverables might have a low degree of change and low frequency of delivery such as weekly progress reports, a high degree of change and a high frequency of delivery such as certain software features, and so on.

Below are brief descriptions of several predictive SDLC models.

- The waterfall life cycle model has well-defined, linear stages of systems analysis, design, construction, testing, and support. This life cycle model assumes that requirements will remain stable after they are defined. The waterfall life cycle model is used when risk must be tightly controlled and when changes must be restricted after the requirements are defined. The waterfall approach is used in many large-scale systems projects where complexity and cost are so high that the more rigid steps of the approach help to ensure careful completion of all deliverables. The waterfall approach also makes sense for IT projects that do not involve software development, such as upgrading all the routers and switches in a company to support VoIP phones
- The spiral life cycle model was developed based on refinements of the water- fall model as applied to large government software projects. It recognizes the fact that most software is developed using an iterative or spiral approach rather than a linear approach. The project team is open to changes and revisions later in the project life cycle, and returns to the requirements phase to more carefully clarify and design the revisions. This approach is suitable for projects in which changes can be incorporated with reasonable cost increases or with acceptable time delays. Figure 2-4 illustrates the differences between the waterfall and spiral life cycle models.
- The prototyping life cycle model is used for developing software prototypes to clarify user requirements for operational software. It requires heavy user involvement, and developers use a model to generate functional requirements and physical design specifications simultaneously. Developers can throw away

or keep prototypes, depending on the project. This approach is often used in systems that involve a great deal of user interface design, such as website projects, in systems that automate previously manual functions, or in systems that change the nature of how something is done, such as mobile applications.

- The RAD life cycle model uses an approach in which developers work with an evolving prototype. This life cycle model also requires heavy user involvement and helps produce systems quickly without sacrificing quality. Developers use RAD tools such as CASE (computer-aided software engineering), JRP (joint requirements planning), and JAD (joint application design) to facilitate rapid prototyping and code generation. These tools are often used in reporting

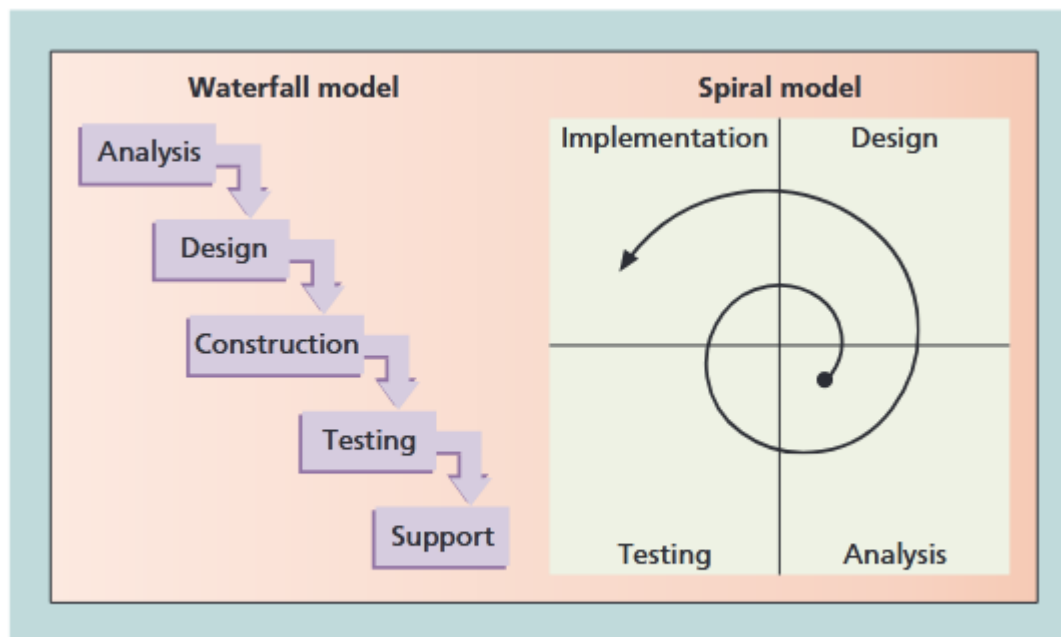


FIGURE 2-4 Waterfall and spiral life cycle models

PROJECT IMPLEMENTATION

Implementation is the realization of an application, execution of a plan, idea, model, design, specification, standard, algorithm, policy, or the administration or management of a process or objective. The term 'implement' means to "put into effect" or "carry out." In project management, to implement a project means executing tasks according to the predefined plan.

Project implementation is the phase where a project plan is put into action, turning strategies and designs into real deliverables through coordinated tasks, resource management, and progress monitoring to achieve objectives within scope, time, and budget. It bridges planning and completion, requiring clear communication, risk management, and performance measurement to ensure tasks align with goals, stakeholders are involved, and unexpected issues are handled effectively.

Topic Four:

PROJECT IMPLEMENTATION

Project implementation is a critical phase in project management where plans turn into reality. This stage involves executing the strategies, deploying resources, and ensuring that objectives are met efficiently. Without a proper implementation plan, even the best project ideas can fail.

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Project implementation refers to the process of executing planned activities to achieve a project's objectives. This phase includes mobilizing resources, managing tasks, and monitoring progress to ensure successful project completion.

STEPS IN PROJECT IMPLEMENTATION

1. Planning the Implementation Process

A well-structured implementation plan is the foundation of a successful project. This includes:

- ✓ Defining clear goals
- ✓ Allocating resources
- ✓ Setting timelines
- ✓ Establishing risk management strategies

2. Assigning Responsibilities

An implementation project manager plays a crucial role in ensuring smooth execution. They delegate tasks, monitor progress, and ensure project milestones are met.

3. Deploying Resources

The next step is mobilizing human, financial, and technological resources needed for the project. This ensures that teams have the necessary tools to perform their tasks.

4. Executing the Plan

Now comes the action phase—where tasks are carried out as per the plan. The team follows predefined processes while adapting to any necessary changes.

5. Monitoring and Controlling

Implementation is not just about execution; it's also about tracking progress. This includes:

- ✓ Regular status updates
- ✓ Performance assessments
- ✓ Adjustments based on feedback

6. Closing the Project

Once all tasks have been implemented, a final evaluation takes place. The team ensures that all objectives have been met before formally closing the project.

The Role of an Implementation Project Manager

An implementation project manager is responsible for overseeing the execution of a project. Their duties include:

- **Planning & Strategy:** Developing the project implementation roadmap.
- **Team Coordination:** Assigning roles and ensuring smooth communication.
- **Problem-Solving:** Addressing challenges that arise during execution.
- **Performance Monitoring:** Evaluating progress to keep the project on track.

A skilled implementation project manager ensures that project goals are met within scope, time, and budget.

Organizational Transformation Implementation Project

Organizational transformation implementation projects involve large-scale changes, such as restructuring, digital transformation, or process improvements. Successful implementation requires:

- Strong leadership

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- Effective change management strategies
- Employee engagement
- Continuous monitoring and adaptation

An example of an organizational transformation implementation project is a company shifting from traditional marketing to digital marketing. This requires new tools, strategies, and staff training, all of which need proper implementation to succeed.

CHALLENGES IN PROJECT IMPLEMENTATION

Even with a solid plan, several challenges may arise during project implementation, such as:

- **Scope Creep:** Uncontrolled expansion of project requirements
- **Resource Constraints:** Lack of funds, time, or workforce
- **Resistance to Change:** Employees resisting new processes or technologies
- **Communication Gaps:** Poor coordination among teams

Topic Five:

RISK MANAGEMENT PROCESS IN PROJECT MANAGEMENT

Project risk management is the process of identifying, analyzing and responding to any risk that arises over the life cycle of a project to help the project remain on track and meet its goal. Risk management isn't reactive only; it should be part of the planning process to figure out the risk that might happen in the project and how to control that risk if it in fact occurs.

A risk is anything that could potentially impact your project's timeline, performance or budget. Risks are potentialities, and in a project management context, if they become realities, they then become classified as "issues" that must be addressed with a risk response plan. So risk management, then, is the process of identifying, categorizing, prioritizing and planning for risks before they become issues.

Risk management can mean different things on different types of projects. On large-scale projects, risk management strategies might include extensive detailed planning for each risk to ensure mitigation strategies are in place if project issues arise. For smaller projects, risk management might mean a simple, prioritized list of high, medium and low-priority risks.

WHAT ARE POSITIVE RISKS & NEGATIVE RISKS IN PROJECT MANAGEMENT?

Risk can be either positive or negative, though most people assume risks are inherently the latter. Where negative risk implies something unwanted that has the potential to irreparably damage a project, positive risks are opportunities that can affect the project in beneficial ways.

Negative risks are part of your risk management plan, just as positive risks should be, but the difference is in approach. You manage and account for known negative risks to neutralize their impact, but positive risks can also be managed to take full advantage of them.

There are many examples of positive risks in projects: you could complete the project early, you could acquire more customers than you accounted for, you could imagine how a delay in shipping might open up a potential window for better marketing opportunities, etc. It's important to note, though, that these definitions are not etched in stone. Positive risk can quickly turn to negative risk and vice versa, so you must be sure to plan for all eventualities with your team.

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RISK MANAGEMENT STATISTICS AND TRENDS

A large factor may be due to the digital transformation of organizations. More companies are adopting advanced analytics, machine learning and AI to improve their risk management processes. As technology increases, the risk of cybercrime increases. The global cost of cybercrime is expected to reach \$13.28 trillion by 2028.

As a result of these growing statistics, the role of a risk manager is also increasing in importance. This is the direct result of factors such as the need for proactive risk management strategies, increased risk complexity in the workplace and a general sense of interconnectedness between people and technology.

6 STEPS IN THE RISK MANAGEMENT PROCESS

So, how do you handle something as seemingly elusive as project risk management? You make a risk management plan. It's all about the process. Turn disadvantages into an advantage by following these six steps.

1. Identify the Risk

You can't resolve a risk if you don't know what it is. There are many ways to identify risk. As you do go through this step, you'll want to collect the data in a risk register.

One way is brainstorming with your team, colleagues or stakeholders. Find the individuals with relevant experience and set up interviews so you can gather the information you'll need to both identify and resolve the risks. Think of the many things that can go wrong. Note them. Do the same with historical data on past projects. Now your list of potential risks has grown.

Make sure the risks are rooted in the cause of a problem. Basically, drill down to the root cause to see if the risk is one that will have the kind of impact on your project that needs identifying. When trying to minimize risk, it's good to trust your intuition. This can point you to unlikely scenarios that you just assume couldn't happen. Use a risk breakdown structure process to weed out risks from non-risks.

2. Analyze the Risk

Analyzing risk is hard. There is never enough information you can gather. Of course, a lot of that data is complex, but most industries have best practices, which can help you with your risk analysis. You might be surprised to discover that your company already has a framework for this process.

When you assess project risk, you can ultimately and proactively address many impacts, such as avoiding potential litigation, addressing regulatory issues, complying with new legislation, reducing your exposure and minimizing impact.

So, how do you analyze risk in your project? Through qualitative and quantitative risk analysis, you can determine how the risk is going to impact your schedule and budget.

Project management software helps you analyze risk by monitoring your project. Project Manager takes that one step further with real-time dashboards that display live data. Unlike other software tools, you don't have to set up our dashboard. It's ready to give you a high-level view of your project from the get-go. We calculate the live date and then display it for you in easy-to-read graphs and charts. Catch issues faster as you monitor time, costs and more.

3. *Prioritize Risks & Issues*

Not all risks are created equally. You need to evaluate the risk to know what resources you're going to assemble towards resolving it when and if it occurs.

Having a large list of risks can be daunting. But you can manage this by simply categorizing risks as high, medium or low. Now there's a horizon line and you can see the risk in context. With this perspective, you can begin to plan for how and when you'll address these risks. Then, if risks become issues, it's advisable to keep an issue log so you can keep track of each of them and implement corrective actions.

Some risks are going to require immediate attention. These are the risks that can derail your project. Failure isn't an option. Other risks are important, but perhaps do not threaten the success of your project. You can act accordingly. Then there are those risks that have little to no impact on the overall project's schedule and budget. Some of these low-priority risks might be important, but not enough to waste time on.

4. *Assign an Owner to the Risk*

All your hard work identifying and evaluating risk is for naught if you don't assign someone to oversee the risk. In fact, this is something that you should do when listing the risks. Who is the person who is responsible for that risk, identifying it when and if it should occur and then leading the work toward resolving it?

That determination is up to you. There might be a team member who is more skilled or experienced in the risk. Then that person should lead the charge to resolve it. Or it might just be an arbitrary choice. Of course, it's better to assign the task to the right person, but equally important in making sure that every risk has a person responsible for it.

Think about it. If you don't give each risk a person tasked with watching out for it, and then dealing with resolving it when and if it should arise, you're opening yourself up to more risk. It's one thing to identify risk, but if you don't manage it, then you're not protecting the project.

5. *Respond to the Risk*

Now the rubber hits the road. You've found a risk. All that planning you've done is going to be put to use. First, you need to know if this is a positive or negative risk. Is it something you could exploit for the betterment of the project? If not you need to deploy a risk mitigation strategy.

A risk mitigation strategy is simply a contingency plan to minimize the impact of a project risk. You then act on the risk by how you prioritize it. You have communications with the risk owner and, together, decide on which of the plans you created to implement to resolve the risk.

6. *Monitor the Risk*

You can't just set forces against risk without tracking the progress of that initiative. That's where the monitoring comes in. Whoever owns the risk will be responsible for tracking its progress toward resolution. However, you'll need to stay updated to have an accurate picture of the project's overall progress to identify and monitor new risks.

Whatever you choose to do, remember to always be transparent. It's best if everyone in the project knows what is going on, so they know what to be on the lookout for and help manage the process.

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In the video below, Jennifer Bridges, professional project manager (PMP) dives deeper into the steps in the risk management process.

EXAMPLES OF RISKS & RISK MANAGEMENT STRATEGIES

1. Construction Risk Management

Construction projects are inherently risky. Numerous potential hazards can impact safety, timelines and budgets. Construction risks may be related to finances, safety, schedules, environment, design or contracts.

The risk: If there are material delivery issues, design changes, labor shortages or bad weather, it may result in a schedule risk. This uses additional resources and can delay the timeline.

Risk management strategy: One example of a construction risk management strategy is contingency planning and creating a schedule with realistic buffer time. Project managers can use critical path analysis to identify key dependencies and monitor the schedule for any updates. If there are potential delays, a contingency plan can help mitigate them.

2. Manufacturing Risk Management

Manufacturing projects range from setting up a new production line to launching a new product. Risk management can help minimize disruptions and improve profitability.

The risk: Production risks such as equipment failures, production line downtime or quality control issues can cause the project to be delivered late or at a lower quality than expected.

Risk management strategy: Quality control procedures such as implementing preventive maintenance programs can help ensure that production risks don't disrupt the process. This is an example of a risk mitigation strategy. Employees should receive ongoing training and development to help spot equipment failures before they snowball into something more serious.

3. IT Risk Management

IT projects are known for their complexity and rapid pace. Risk management in IT can help maximize the return on investment and minimize disruptions.

The risk: Technical risks such as integration issues, technology failures or performance problems may result in throwing the project off schedule and using more resources than originally planned.

Risk management strategy: To avoid technical risks, the IT project manager can implement contingency planning such as thorough technology assessments and testing before the project kicks off. They may also utilize industry best practices and expert technical resources to ensure any performance issues are addressed as quickly as possible.

4. Marketing Risk Management

From rebranding a product to launching a new campaign, marketing projects are subject to risks that can impact their success.

The risk: There's a new creative marketing campaign being launched that has the potential to fail to resonate with the target audience. This would waste significant time and resources.

Risk management strategy: To avoid creative concepts failing, the marketing project manager and team can conduct rigorous creative testing and feedback loops to ensure they understand their audience. If changes are needed before launch, the team can implement the strategy accordingly.

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LEADERSHIP AND ETHICS IN PROJECTS

Leadership and ethics in projects are crucial for success, involving guiding principles like honesty, responsibility, fairness, and respect to build trust, ensure transparency, and foster a positive environment, with ethical leaders modeling integrity to inspire teams, make sound decisions, and navigate dilemmas, ultimately leading to better project outcomes and a stronger organizational culture.

WHAT IS ETHICAL LEADERSHIP AND WHY IS IT IMPORTANT?

Ethical leadership involves leaders and managers making decisions based on the right thing to do for the common good, not just based on what is best for themselves or for the bottom line.

While profits are important, ethical leaders take into consideration the needs of customers, communities, and employees in addition to company growth and revenue when making business decisions.

Ethical leaders encourage their team members to model this behavior, too. They help to build a workplace culture that values transparency, collaboration and inclusion, and where everyone feels safe to share their voice.

They can also help organizations recruit and retain top talent. Professionals are increasingly seeking out companies whose leaders strive to do the right thing. generation Z, who will make up 25 percent of the workforce by 2025, demands leadership ethics more than generations that came before them.

“Gen Z is not going to negotiate. They have really strong values and ethics, and they don’t bend them because of intimidation or because they are just getting a paycheck,” said Michael McCarthy, instructor at Harvard Division of Continuing Education’s Professional & Executive Development and host of the “Happy at Work” podcast. “The idea of letting harmful or hurtful behavior slide is not acceptable.”

Leaders who weigh ethical considerations before making key business decisions drive a company’s long-term success.

THE 6 MAIN PRINCIPLES OF ETHICAL LEADERSHIP

1. Respect

Respect includes valuing others’ skills and contributions. While historically respect in the workplace may have been one-way (leaders demanding respect from employees), in an ethical work environment, respect is mutual.

Mutual respect leads to healthier workplace relationships where both sides appreciate and support what the other is doing and feel secure in talking through issues and challenges. Healthy relationships create positive work environments, which drives increased productivity.

Current and upcoming business leaders should take mutual respect into account as workforce expectations continue to shift.

“I tell current leadership to respect Gen Z. They have values and morals, and you’re going to have a better organization because of them,” McCarthy said. “They aren’t going to put up with the old hierarchy that doesn’t offer mutual respect.”

2. Accountability

Ethical leaders hold themselves accountable for their actions. They make decisions based on integrity and stand behind their work. They also lead by example, communicate openly about challenges, and don't look to place blame on others for any shortfalls.

3. Service

Leaders make ethical decisions based on doing what is right for employees, customers, and the community. Because these constituents are always top of mind for ethical leaders, they often have a strong sense of service. They engage in activities such as charitable giving and volunteer work to give back to their communities — and encourage their teams to do the same.

4. Honesty

Leaders who are transparent build trust amongst their organizations and amongst customers. To build and maintain trust, leaders must be good communicators who speak openly and honestly about issues. Regardless of the issue's severity or unpopularity, leaders' responsibility to be clear and candid empowers others to make the right decisions with the information they have. Honesty and transparency also help to build a brand's reputation, leading to long-term customer loyalty.

5. Justice

Justice is not just about following the law, but about ensuring that everyone is getting what they deserve. Ethical leaders approach situations with a focus on treating everyone fairly, and they expect their teams to treat each other and customers the same way. Through their actions, they build equitable work environments where everyone feels respected.

6. Community

Ethical leaders view their companies as communities and consider everyone involved when evaluating situations and making decisions. By viewing their organizations this way, they build equity and inclusion into their decision-making process and create work environments that encourage collaboration across teams.

Topic Six:

PROJECT QUALITY MANAGEMENT

The purpose of project quality management is to ensure that the project will satisfy the needs for which it was undertaken. Project quality management involves three main processes:

1. Planning quality management includes identifying which quality requirements and standards are relevant to the project and how to satisfy them. Incorporating quality standards into project design is a key part of quality planning. For an IT project, quality standards might include allowing for system growth, planning a reasonable response time for a system, or ensuring that the system produces consistent and accurate information. Quality standards can also apply to IT services. For example, you can set standards for how long it should take to get a reply from a help desk or how long it should take to ship a replacement part for a hardware item under warranty. The main outputs of planning quality management are a quality management plan, quality metrics, project management plan updates, and project documents updates. A metric is a standard of measurement. Examples of common metrics include failure rates of products, availability of goods and services, and customer satisfaction ratings.

2. Managing quality involves translating the quality management plan into executable quality activities. These activities must adhere to the organization's quality policies. The main outputs of this process are

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quality reports, test and evaluation documents, change requests, project management plan updates, and project documents updates.

3. Controlling quality involves monitoring specific project results to ensure that they are complete, correct, and meet customer expectations. This process is often associated with the technical tools and techniques of quality management, such as Pareto charts, quality control charts, and statistical sampling.

You will learn more about these tools and techniques later in this chapter. The main outputs of quality control include quality control measurements, verified deliverables, work performance information, change requests, project management plan updates, and project documents updates

Topic Seven:

PROJECT SCHEDULING

Project scheduling is the process of mapping out a project's tasks, resources, and timelines to create a detailed roadmap for successful, on-time delivery, involving defining activities, setting durations, identifying dependencies, and allocating people/materials, often visualized with tools like Gantt charts to track progress and manage issues like bottlenecks. It's crucial for communication, resource allocation, and ensuring goals align with deadlines.

Project scheduling techniques are essential tools for organizing, planning and managing tasks efficiently.

Key Components of a Project Schedule

Tasks/Activities: The specific work to be done.

Sequence & Dependencies: The order tasks must be completed in, and which tasks rely on others.

Duration: Estimated time for each task.

Resources: People, equipment, and budget needed.

Milestones: Key checkpoints or significant events in the project.

Start/End Dates: Planned dates for tasks and the overall project.

Common Scheduling Techniques & Tools

Gantt Charts: Visual bar charts showing task timelines.

Critical Path Method (CPM): Identifies the longest sequence of tasks (the critical path) that determines the project's minimum duration.

PERT Charts: Uses probabilistic time estimates (optimistic, pessimistic, most likely).

Software: Tools like monday.com, Wrike, Asana, Trello help manage complex schedules.

Benefits

Visibility: Provides a clear overview of the entire project.

Communication: Helps everyone understand roles, responsibilities, and timelines.

Proactive Management: Allows managers to spot potential delays and resource conflicts early.

Goal Alignment: Keeps the team focused on delivering on time and within scope.

The Scheduling Process

Define Scope & WBS: Understand the project goals and break work into a Work Breakdown Structure (WBS).

Sequence Activities: Determine dependencies.

Estimate Durations & Resources: Assign time and resources to tasks.

Develop Schedule: Use techniques (like CPM, Gantt) to build the timeline.

Monitor & Control: Track progress, compare actuals to the plan, and make adjustments as needed.

1. **CRITICAL PATH METHOD (CPM)**

The critical path method (CPM) is a project scheduling technique that identifies the longest sequence of dependent tasks. It uses a network diagram and mathematical calculations to estimate the total duration of a project. By determining the critical path, project managers can identify which tasks cannot be delayed without affecting the project deadline. CPM helps highlight float time for non-critical tasks, making it easier to prioritize and reallocate resources when needed.

When to Use This Project Scheduling Technique?

CPM is widely used in industries like construction, engineering and software development, where task dependencies and deadlines are critical. This project scheduling technique is most useful in large-scale projects involving multiple activities with interrelated timelines. It's ideal when you need to manage complex workflows, optimize timelines or analyze the impact of delays. CPM is also effective during project planning and monitoring stages, where detailed time estimates and schedule adjustments are frequently needed.

Benefits of This Project Scheduling Technique

The critical path method offers several advantages that help project managers stay organized and on schedule. Below are some key benefits of using this project scheduling technique and how each contributes to more effective project management.

1. ***Helps Prioritize Tasks:*** CPM highlights which tasks are critical and which have float time, allowing project managers to focus efforts where delays would affect the overall timeline. This clarity improves decision-making and ensures key milestones receive proper attention throughout execution.
2. ***Facilitates Resource Scheduling:*** CPM supports more efficient resource allocation by identifying task dependencies and durations. It allows project managers to assign team members, equipment and materials based on when they're needed most, reducing idle time and improving productivity across the board.
3. ***Helps Accurately Estimate the Duration of Projects Before They Start:*** CPM uses network diagrams and time estimates to calculate the total project duration. This mathematical approach allows for precise scheduling, enabling better forecasting and helping stakeholders set realistic expectations for completion dates and deliverable timelines.

Disadvantages of This Project Scheduling Technique

1. ***Mathematical Calculations Can Be Hard to Grasp:*** CPM requires familiarity with network logic and formulas, which can be difficult for some team members. Misunderstandings or calculation errors can lead to incorrect timelines, poor planning and confusion about task priorities or resource assignments.
2. ***CPM Diagrams Can Become Overly Complex:*** In large projects, the number of tasks and dependencies can make the network diagram overwhelming. Managing updates or analyzing changes becomes time-consuming, especially if the project undergoes frequent adjustments or has unclear task relationships.
3. ***Effectiveness Depends on Accurate Inputs:*** CPM's reliability hinges on the accuracy of task durations and dependencies. If initial data is flawed or overly optimistic, the resulting schedule may be misleading, causing issues with deadlines, budgets and stakeholder trust down the line.

2. PROGRAM EVALUATION AND REVIEW TECHNIQUE (PERT)

The program evaluation and review technique (PERT) is a project scheduling method that helps estimate a project's duration by analyzing the time required to complete each task. It uses a network diagram and mathematical calculations based on optimistic, pessimistic and most likely time estimates. PERT is designed to deal with uncertainty in project timelines and is especially useful in planning and analyzing the time required for tasks with variable durations. It helps improve forecasting and risk assessment in project planning.

When to Use This Project Scheduling Technique?

PERT is most commonly used in research, development and engineering projects where task durations are uncertain or variable. This project scheduling technique is ideal for complex, non-repetitive projects in industries like aerospace, defense and technology. It's especially valuable when you need to forecast project timelines early, even when data is limited or unpredictable. PERT is also helpful during feasibility studies or when project managers want to analyze how different time scenarios affect overall project completion dates.

Benefits of This Project Scheduling Technique

PERT offers a flexible scheduling method ideal for projects with unpredictable elements. Factoring in time variability and focusing on task sequences helps project managers better anticipate risks and plan around uncertainty.

1. ***Attempts to Provide a Realistic Time Estimation for Tasks:*** PERT uses three time estimates—optimistic, pessimistic and most likely—to calculate a weighted average. This provides a more nuanced and realistic forecast than single-point estimates, especially in complex or research-based projects.
2. ***Ideal for Projects with Uncertain Conditions:*** Because it embraces variability, PERT is well-suited for projects with undefined scopes, evolving requirements or new technologies. It allows teams to prepare for delays and better manage scheduling risks from the outset.
3. ***Helps Identify Task Dependencies and Task Sequences:*** PERT charts visually map task relationships and dependencies to help managers spot which tasks influence overall project duration and support smarter sequencing and prioritization decisions.

Disadvantages of This Project Scheduling Technique

While PERT helps with planning under uncertainty, it has limitations in practical execution. It requires manual updates, offers limited real-time control and relies heavily on estimators' assumptions, which can weaken scheduling accuracy.

1. ***Time Estimations Are Subjective:*** PERT depends on team members' ability to guess best-case and worst-case durations. If these estimates are unrealistic or biased, the schedule becomes unreliable and can misguide planning decisions.
2. ***PERT Charts Need to Be Manually Updated:*** Unlike dynamic scheduling tools, PERT diagrams must be revised manually when task durations change or new dependencies arise, making the process time-consuming in active projects.
3. ***Not Ideal for Project Monitoring and Control:*** PERT is more of a planning tool than a tracking system. It lacks mechanisms for real-time updates, progress tracking or integration with cost and resource data, limiting its usefulness during execution.

3. FAST TRACKING

Fast tracking is a schedule compression technique used when a project falls behind and corrective actions are needed to meet deadlines. It involves performing tasks in parallel that were originally planned to be sequential. While it doesn't reduce the scope of work, it helps recover lost time by overlapping activities wherever possible. This technique can accelerate project timelines, but it increases the risk of rework and demands strong coordination to avoid conflicts or quality issues.

When to Use This Project Scheduling Technique?

Fast tracking is commonly used in industries with tight deadlines, such as construction, software development and event planning. It's especially helpful when projects are delayed but the final delivery date must remain unchanged. It suits projects with clearly defined dependencies and teams that can handle overlapping tasks efficiently. Fast tracking is also useful during design-build phases or product launches, where compressing the schedule without cutting scope is critical for meeting regulatory, seasonal or market-driven deadlines.

Below is a fast-tracking example of a residential construction project. Each bar represents a task that must be completed. One task must finish before the rest can begin.

Residential construction project fast tracking example, a project scheduling technique

To fast-track the project, tasks must be executed in parallel. This is possible only for tasks that don't have a dependency, such as wood framing or sheathing installation. However, tasks like roofing work cannot be fast-tracked, as a roof cannot be put on a structure without walls. The image below represents the fast-tracked project with shorter task durations on tasks when possible.

Benefits of This Project Scheduling Technique

Fast tracking is a popular choice for accelerating schedules when time is tight. It helps reduce project duration without increasing resources or changing scope. Below are three key benefits of using this project scheduling technique.

- 1. Accelerates Project Delivery:** Fast tracking shortens timelines by overlapping dependent tasks. Instead of waiting for one task to finish before starting another, teams work in parallel, helping the project finish faster while keeping the full scope intact.
- 2. Avoids Overtime Costs:** Rather than extending work hours, fast tracking compresses the schedule through smarter sequencing. This minimizes the need for overtime or weekend labor, helping reduce additional payroll expenses while catching up on delays.
- 3. No Additional Budget Needed Initially:** Because fast tracking rearranges task order without adding staff or scope, it often doesn't require an immediate increase in budget. It's a time-saving option that can be implemented quickly using existing team capacity.

Disadvantages of This Project Scheduling Technique

While fast-tracking helps speed up projects, it has trade-offs that can affect quality, resources and overall coordination. Here are some common drawbacks associated with this technique.

Increases Risk of Rework: Overlapping tasks can lead to rework if earlier tasks are revised after their successors have already started. This adds unexpected delays and costs, potentially canceling the original time savings.

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May Strain Resources: Fast tracking can overload teams or equipment by requiring multiple activities to run simultaneously. Without careful resource planning, it may result in burnout, bottlenecks or unproductive downtime.

Not Suitable for All Tasks: Some tasks have hard dependencies and can't be performed in parallel. Trying to fast-track these may cause confusion, safety risks or compliance issues that harm project execution.

4. PROJECT CRASHING

Project crashing is a schedule compression technique used when a project is delayed and immediate corrective action is required. It involves allocating additional resources—such as labor, equipment, or budget—to tasks on the critical path to reduce their duration. While this method speeds up project completion, it typically increases project costs. Crashing is often used when deadlines are non-negotiable, and other options like fast tracking are not feasible. It requires careful analysis to ensure cost trade-offs are justified.

When to Use This Project Scheduling Technique?

Project crashing is typically used in deadline-driven industries such as construction, engineering, event planning and manufacturing. It's most useful when a project has fallen behind schedule and must meet a fixed deadline due to contractual obligations, regulatory requirements or financial penalties. Crashing is ideal for projects with budget flexibility, clearly defined critical paths and tasks that can benefit from extra labor or resources. It should be considered when speed is more important than maintaining the original cost baseline.

Benefits of This Project Scheduling Technique

Project crashing can be a useful method to recover lost time. It allows teams to meet tight deadlines without changing project scope, especially when schedule is a higher priority than cost.

1. ***Recovers Delayed Schedules Quickly:*** Crashing shortens the overall project duration by adding resources to critical path tasks. It's particularly effective when deadlines are at risk and no float remains, allowing project teams to regain lost time without reducing the scope of work.
2. ***Maintains Project Scope:*** Unlike other time-saving strategies that cut deliverables, crashing keeps the full scope intact. Teams can meet fixed deadlines while maintaining all project requirements, making it ideal for high-visibility or regulated projects where scope reduction would compromise quality or contractual compliance.
3. ***Protects Against Late Penalties:*** When delays trigger financial penalties, legal consequences, or client dissatisfaction, crashing becomes a useful tool. By accelerating critical tasks, teams can avoid costly overruns, fulfill contractual deadlines, and preserve client relationships without altering scope or sacrificing quality deliverables.

Disadvantages of This Project Scheduling Technique

While project crashing can save time, it has trade-offs. The method often increases costs, puts pressure on resources and is only useful for tasks on the critical path.

1. ***Increases Project Costs Significantly:*** Crashing often requires additional labor, expedited materials, or specialized equipment to reduce task durations. These accelerated efforts significantly increase the project's overall cost, making crashing a financially intensive option that should only be used when the deadline justifies the extra investment.
2. ***Can Lead to Team Burnout:*** Overloading existing team members or rapidly onboarding new staff to meet compressed timelines can lead to burnout, confusion or communication issues. If not

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properly managed, these pressures may reduce productivity, lower morale and negatively affect long-term team performance.

3. ***Not Always Effective on All Tasks:*** Project crashing only impacts tasks on the critical path. Applying it to non-critical activities doesn't reduce the project duration and wastes resources. Therefore, it must be applied selectively and strategically to be cost-effective and schedule-relevant.

5. MILESTONE CHARTS

Milestone charts are visual project scheduling tools that highlight key events or deliverables along a timeline. Unlike detailed Gantt charts, they simplify complex schedules by focusing only on major checkpoints. This makes them ideal for presenting project progress to stakeholders or executives who need a high-level overview. Milestone charts don't show task durations or dependencies but offer clarity by tracking critical phases, approvals and deadlines that define overall project progress.

When to Use This Project Scheduling Technique?

Milestone charts are best used in projects where stakeholders need quick, high-level updates. They're common in industries like construction, product development, and consulting, especially when executive reporting or client presentations are involved. This technique is useful during the early planning phase, for phase-gate reviews, or to track major deliverables in long-term projects. It's ideal when full task-level detail is unnecessary, but visibility into major deadlines is critical.

Benefits of This Project Scheduling Technique

Milestone charts offer a clean, simplified view of project progress. They're especially helpful when reporting to executives or clients who need clarity without the clutter of task-level details.

1. ***Simplifies Complex Project Timelines:*** Milestone charts focus only on major events, helping teams and stakeholders understand project progress at a glance without diving into every task or dependency. This streamlines communication and reduces confusion in complex or large-scale projects.
2. ***Improves Stakeholder Communication:*** These charts are ideal for status reporting. Their simplicity makes it easy for non-technical stakeholders to track deadlines and phase completions, enabling clearer conversations around project expectations, timing and progress.
3. ***Helps Track Key Deliverables:*** Milestone charts keep attention on important goals, approvals and deadlines. They're useful for managing contracts, product launches or gated reviews where meeting specific milestones is critical to project success or payment schedules.

Disadvantages of This Project Scheduling Technique

1. ***Despite their clarity, milestone charts have limitations.*** They don't show how tasks connect, how long they take or what happens between checkpoints—so they're best used alongside more detailed scheduling tools.
2. ***Lacks Task-Level Detail:*** Milestone charts don't display individual tasks, durations or dependencies. That makes them ineffective for managing daily project work, assigning resources, or tracking incremental progress between major deliverables.
3. ***Not Suitable for Complex Execution:*** While useful for reporting, milestone charts can't guide detailed scheduling decisions. Without visibility into task sequences or resource needs, they can't support day-to-day project management or execution planning.
4. ***Can Oversimplify Project Status:*** By omitting task-level data, milestone charts may give a false sense of progress. A milestone might appear "on track" even if the tasks leading to it are delayed or at risk.

6. GANTT CHARTS

A Gantt chart is one of the most powerful and widely used tools in project scheduling. It provides a visual timeline of tasks, showing their start and end dates, durations, and progress. Gantt charts allow project managers to link task dependencies, so if one task is delayed, related tasks shift accordingly. They can also identify the critical path, helping teams focus on the activities that directly affect the project's end date.

In addition to scheduling, Gantt charts support resource management by assigning team members or equipment to specific tasks. Many modern Gantt chart tools also track costs, monitor task progress in real time and adjust timelines based on changes. This makes it possible to integrate time, scope, cost, and resource data into a single visual interface. The result is a smarter, more centralized way to manage complex projects from planning through execution.

When to Use This Project Scheduling Technique?

Gantt charts are ideal for projects with many tasks, dependencies, and deadlines. They're commonly used in industries like construction, IT, product development, and marketing. This technique is especially useful for managing cross-functional teams, coordinating resources, and tracking progress across multiple work streams. Use Gantt charts when visual clarity, timeline accuracy, and detailed planning are essential—particularly in long-term or phased projects that require transparency, frequent reporting, and adaptive rescheduling to keep all stakeholders aligned.

Benefits of This Project Scheduling Technique

Gantt charts are effective because they centralize planning elements in one view. They allow project managers to visualize progress, adjust in real time and communicate clearly with all stakeholders.

1. **Connects Task Dependencies Visually:** Gantt charts clearly show how tasks are linked, helping teams understand which activities must be completed before others begin. This improves sequencing, prevents bottlenecks and enables better planning of start and end dates.
2. **Monitors Progress in Real Time:** By updating task bars as work is completed, Gantt charts offer a live view of project progress. This helps managers spot delays quickly and make informed decisions to get back on schedule.
3. **Combines Scope, Time and Resources:** Modern Gantt tools integrate deadlines, assigned team members, budgets and milestones all in one chart. This unified view makes it easier to manage the entire project efficiently without switching between multiple tools.

Disadvantages of This Project Scheduling Technique

While Gantt charts are powerful scheduling tools, they can become overly complex in large projects. Their visual clarity can diminish as task volume increases, and they require consistent updates to remain accurate and useful for ongoing project execution.

1. **Can Become Cluttered in Large Projects:** With too many tasks and dependencies, a Gantt chart can turn into a tangled web that's hard to read. This limits its effectiveness in programs with hundreds of activities or frequent task changes.
2. **Requires Frequent Manual Updates:** Without automation, Gantt charts need regular updating to reflect schedule changes, resource shifts or progress. If not maintained properly, they become outdated quickly and can mislead stakeholders.

- 3. Lacks Detail for Day-to-Day Execution:** While good for big-picture planning, Gantt charts don't always show task-level nuances, such as time spent, communication or small blockers. This makes them insufficient as a standalone tool for managing execution in complex, fast-moving projects.

7. PRECEDENCE DIAGRAMMING METHOD (PDM)

The precedence diagramming method (PDM) is a visual scheduling technique used in project Management to represent task relationships. It shows the logical sequence of activities using nodes and arrows, where each node is a task and arrows indicate dependencies. PDM supports four dependency types: finish-to-start, start-to-start, finish-to-finish and start-to-finish. This allows project managers to build accurate, flexible project schedules and identify the critical path. PDM is widely supported in modern scheduling tools like Gantt charts.

When to Use This Project Scheduling Technique?

PDM is best used in complex projects where task relationships must be clearly defined. It's widely applied in construction, engineering, IT, and manufacturing. This method is especially useful in projects with overlapping or interdependent activities that require accurate sequencing. Use PDM when multiple dependency types must be managed or when critical path identification is essential for timeline control. It's ideal for teams using scheduling software and working with strict deadlines, scope requirements or formal project methodologies.

Benefits of This Project Scheduling Technique

PDM helps create detailed, logic-based schedules that improve accuracy and control. It's especially helpful when managing large projects with multiple dependency types and complex timelines.

- 1. Clarifies Task Relationships:** PDM shows how activities connect through visual nodes and arrows, helping teams understand the order of operations. This clarity supports better planning, sequencing, and coordination across all project phases.
- 2. Supports Multiple Dependency Types:** Unlike simpler methods, PDM accounts for various relationships, like start-to-start or finish-to-finish. This flexibility makes scheduling more realistic, especially in projects where tasks overlap or need to start concurrently.
- 3. Enables Critical Path Analysis:** By mapping out dependencies, PDM helps identify the critical path—the sequence of tasks that directly affect the project's end date. This allows project managers to focus on key activities and manage risks effectively.

Disadvantages of This Project Scheduling Technique

Despite its advantages, PDM can be resource-intensive and may not suit every project. It's best used with proper tools and trained teams who can handle the level of detail required.

- 1. Time-Consuming to Create:** Building a PDM network requires detailed input on task relationships, durations and logic. This can be time-intensive during the planning phase, especially for large projects with many interdependent activities.
- 2. Requires Specialized Knowledge:** To use PDM effectively, teams must understand dependency logic and be comfortable working with scheduling software. Without training, users may misrepresent task flows or overlook important constraints.
- 3. Less Useful for Simple Projects:** For small or straightforward projects, PDM may be overkill. The complexity of creating and maintaining a diagram outweighs its benefits when only a handful of tasks or dependencies are involved.

8. ARROW DIAGRAMMING METHOD (ADM)

The arrow diagramming method (ADM), also known as the activity-on-arrow (AOA) method, is a project scheduling technique where arrows represent activities and nodes indicate events or milestones. ADM is used to visually sequence tasks and define dependencies between them. It's one of the earliest scheduling methods and helps identify the critical path by showing task order and duration.

When to Use This Project Scheduling Technique?

ADM is best used in projects that follow traditional or linear workflows, such as civil engineering, infrastructure, or manufacturing. It's suited for teams working with paper-based or legacy systems and for educational purposes when teaching critical path analysis. ADM is helpful when projects require clear milestone tracking and when activities follow a straightforward sequence. While it's less flexible than modern methods, ADM remains a valid choice for structured, scope-driven projects with well-defined task durations.

Benefits of This Project Scheduling Technique

ADM can still offer value for teams looking to visualize activity flow in simple, milestone-driven projects. It's especially helpful in planning environments that emphasize clarity over flexibility.

1. ***Visualizes Project Milestones Clearly:*** ADM emphasizes key events by representing them as nodes. This helps stakeholders and teams understand when major points in the project must be reached and how activities connect between those milestones.
2. ***Simplifies Task Sequences:*** By focusing on activity order rather than detailed task attributes, ADM makes it easier to visualize sequential workflows—ideal for projects with predictable, step-by-step execution.
3. ***Facilitates Basic Critical Path Analysis:*** ADM supports identifying the critical path in a clear and logical format. This makes it useful for teaching scheduling fundamentals or managing smaller projects with limited interdependencies.

Disadvantages of This Project Scheduling Technique

While ADM has historical significance, it comes with several limitations that make it less practical for complex or dynamic project environments.

1. ***Limited Dependency Types:*** ADM only supports finish-to-start relationships. It lacks the flexibility of methods like PDM, which allow for more nuanced dependency modeling, such as start-to-start or finish-to-finish.
2. ***Can Require Dummy Activities:*** To represent complex task logic, ADM often needs dummy activities—tasks that take no time but must be shown for logic's sake. These can clutter diagrams and confuse stakeholders.
3. ***Outdated in Modern Software Tools:*** Most project management platforms now favor PDM. ADM is rarely supported in newer scheduling tools, which can make implementation cumbersome or limit integration with other planning features.

9. ROLLING WAVE PLANNING

Rolling wave planning is a project scheduling technique that involves planning work in greater detail as the project progresses. It's a form of progressive elaboration, where near-term tasks are defined in detail, while future work is kept at a high level until more information becomes available. This allows project teams to adapt to changing conditions, reduce planning rework and maintain flexibility. Rolling wave planning is especially useful in dynamic or evolving projects where detailed foresight isn't fully possible at the start.

When to Use This Project Scheduling Technique?

Rolling wave planning is ideal for projects with uncertain scopes, changing requirements, or iterative phases. It's frequently used in software development, R&D, or innovation-driven projects where details emerge over time. It also benefits large infrastructure projects planned in phases. Use it when early milestones are known, but later activities require refinement. This technique works well in Agile or hybrid project environments, where flexibility and ongoing updates to the project schedule are critical to success.

Benefits of Using This Project Scheduling Technique

Rolling wave planning offers flexibility in uncertain environments. It allows for early execution while giving teams time to develop accurate plans for future tasks.

1. ***Improves Planning Accuracy Over Time:*** Instead of guessing details too early, rolling wave planning enables more accurate task definitions when information becomes available. This reduces rework and leads to better resource, cost and duration estimates.
2. ***Supports Flexibility in Dynamic Projects:*** In rapidly changing projects, this method allows teams to adjust timelines and plans without constant major overhauls. It keeps the overall framework stable while adapting to real-time inputs.
3. ***Enables Early Project Execution:*** Work can begin on well-understood tasks even while future phases are being planned. This shortens lead times and prevents delays while still allowing room for strategic thinking on long-term goals.

Disadvantages of This Project Scheduling Technique

While useful for adaptive planning, rolling wave planning can introduce uncertainty and complexity. It requires strong oversight and ongoing updates to stay effective.

1. ***Lack of Detail Can Confuse Stakeholders:*** Future work remains vague until later phases, which can lead to misunderstanding or concern among stakeholders who expect a complete plan from the start.
2. ***Demands Ongoing Planning Effort:*** Rolling wave planning isn't a one-time activity. It requires repeated planning sessions, timeline adjustments and team alignment as the project moves forward.
3. ***Challenging to Estimate Overall Timeline:*** Because later phases aren't fully defined, it can be difficult to accurately estimate the full project duration or forecast long-term resource needs early on.

10. RESOURCE LEVELING

Resource leveling is a project scheduling technique used to resolve resource overallocations by adjusting start and end dates based on availability. When resources such as labor, equipment or budget are overbooked or spread too thin, this method balances workloads across the schedule. Tasks may be delayed or extended, but overallocation is avoided. Resource leveling helps project managers create more realistic timelines and avoid burnout or missed deadlines due to resource shortages, making it essential for long-term planning and execution.

When to Use This Project Scheduling Technique?

Resource leveling is most useful when project resources are limited or shared across multiple initiatives. It's common in construction, IT, consulting, and manufacturing—anywhere resource bottlenecks occur. Use it when team members are overallocated, or when delays are preferable to overworking staff or adding costs. It's also helpful in portfolio management, where multiple projects compete for the same resources. Leveling is ideal when maintaining quality and avoiding burnout are more important than finishing the project as fast as possible.

Benefits of Using This Project Scheduling Technique

Resource leveling improves schedule realism and helps teams avoid overload. It's especially beneficial in environments with limited personnel or shared equipment.

1. ***Reduces Resource Overload:*** By balancing resource demand, this method prevents overuse of individuals or equipment. That reduces the risk of burnout, inefficiency or mistakes caused by excessive multitasking or unrealistic workloads.
2. ***Improves Timeline Accuracy:*** Leveling aligns project schedules with real-world resource limits, making completion dates more achievable. This leads to better planning, stakeholder trust and fewer last-minute adjustments caused by staffing issues.
3. ***Promotes Sustainable Team Performance:*** Spreading workload more evenly helps preserve morale, engagement and productivity over time. Teams are less likely to experience stress or turnover due to extended overcommitment.

Disadvantages of This Project Scheduling Technique

While resource leveling solves allocation problems, it can extend project timelines and complicate sequencing. It requires careful planning and may not be ideal when speed is the top priority.

1. ***Can Delay Project Completion:*** Leveling often requires shifting task start dates to accommodate resource availability, which may push the overall project end date further out.
2. ***Increases Schedule Complexity:*** Adjusting for resource limits introduces new dependencies or timing gaps that complicate planning. The schedule becomes more difficult to visualize and manage.
3. ***Not Effective in All Project Types:*** In highly deadline-driven projects, such as regulatory or event-based work, delays may be unacceptable. In such cases, leveling may conflict with time constraints.

Possible Questions

1. Why is there a new or renewed interest in the field of project management?
 2. What is a project, and what are its main attributes? How is a project different from what most people do in their day-to-day jobs? What is the triple constraint? What other factors affect a project?
 3. What is project management? Briefly describe the project management framework, providing examples of stakeholders, knowledge areas, tools and techniques, and project success factors.
 4. What is a program? What is a project portfolio? Discuss the relationship between projects, programs, and portfolio management and the contributions that each makes to enterprise success.
 5. What is the role of the project manager? What are suggested skills for all project managers and for IT project managers? Why is leadership so important for project managers? How is the job market for IT project managers?
 6. Briefly describe some key events in the history of project management. What role do the Project Management Institute and other professional societies play in helping the profession?
 7. What functions can you perform with project management software? What are the main differences between low-end, midrange, and high-end project management tools?
 8. Discuss ethical decisions that project managers often face. Do you think a professional code of ethics makes it easier to work in an ethical manner?
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1. ***Which of the following is not a potential advantage of using good project management?***
 - a) Shorter development times
 - b) Higher worker morale
 - c) Lower cost of capital
 - d) Higher profit margin

2. A _____ is a temporary endeavor undertaken to create a unique product, service, or result.

- a) program
- b) process
- c) project
- d) portfolio

3. Which of the following is not an attribute of a project?

- a) Projects are unique.
- b) Projects are developed using progressive elaboration.
- c) Projects have a primary customer or sponsor.
- d) Projects involve little uncertainty.

4. _____ is the application of knowledge, skills, tools, and techniques to project activities to meet project requirements.

- a) Project management
- b) Program management
- c) Project portfolio management
- d) Requirements management

Questions

1. Explain the term 'Project Management'.
2. List the core components of Project Management.
3. Explain the Critical Path Method (CPM).
4. Explain the Programme Evaluation & Review Technique (PERT).
5. Differentiate between CPM & PERT.
6. Explain the Concept of Break-Even-Analysis (BEA).
7. Construct a network diagram.
8. What are the elements of project formulation?
9. List the stages of project management.